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SUBSTATION CLEARANCES

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TRANSMISSION ENGINEERING STANDARD

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1.0 SCOPE & PURPOSE

1.1 This Engineering Standard indicates the principles and concepts which are bases for selecting various air clearances for substations in the electrical system of National Grid Saudi Arabia.

1.2 The purpose is to specify the clearances which shall be applicable within the substation / substation bays for rigid conductors and short lengths of stain conductors in field installations, that cannot be impulse tested, as whole. These clearances are not applicable to equipment and apparatus, which have been successfully type tested and guaranteed by the equipment manufacturer.

1.3 The ground and phase clearances between factory tested equipment / apparatus and other equipment / structures / buildings shall be as specified in this standard.

2.0 GENERAL GUIDELINES

2.1 In the installations such as substations, which cannot be tested as a whole, it is necessary to ensure that the dielectric strength of air gaps is adequate.

2.2 The dielectric strength of air gaps between phase conductors or rigid busbars and grounded metal frames under lightning surge and switching surge impulse voltages determines the required air clearances between them.

2.3 Space requirements and layout of electrical equipment in the substations also depend upon various types of electrical clearance in air. These air clearances are primarily the phase-to-ground and phase-to-phase safety and maintenance clearances.

2.4 Air clearances indicated in this standard are based on the internal and external insulation levels (BIL and BSL) adopted by National Grid Saudi Arabia as given in TES-P-119.02 and 01-TMSS-01. The clearances are valid for the altitude and site conditions which have been taken into account in selection of these withstand voltages levels.

2.5 Air clearance for voltage levels up to and including 230kV are selected on the basis of the rated lightning surge withstand voltage, i.e. Basic Insulation Level (BIL), since the clearances needed to meet the BIL requirements invariably provide adequate insulation to withstand associated maximum anticipated switching surge voltages. The air clearances for voltages above 230kV are largely dependent on the rated switching surge withstand voltage, i.e. Basic Switching Level (BSL).

2.6 All clearances shall be measured from surface to surface. For clearance measurement, live metallic hardwire electrically connected to a line conductor shall be considered a part of the line conductor. Spacing, where specified, shall be measured center to center. In such a case, it shall be ensured that the minimum metal to metal clearances as specified in Table 08-1 are met.

2.7 The clearance indicated in IEC 60071-2 are intended soley to address in insulation coordination requirements. Safety requirements may result in substantially larger clearances.

2.8 The aspect of insulation coordination and clearances is based on statistical phenomena of over voltage probability density and probability of insulation failure but due to practical difficulty in determining these factors within any degree of accuracy and rare economic incentive for reducing the insulation levels and spacing. The conventional procedure based on an adequate margin is adapted and clearances are not reduced to critical stage. In order to accommodate the contingencies and effect of atmospheric / environmental site conditions like humidity / rain, altitude and contamination severity etc. in National Grid Saudi Arabia transmission system, a margin (found to be adequate from eperience) over the clearances recommended in IEC 60071-2 is considered.

3.0 PHASE – TO- GROUND AIR CLEARANCES

IEC 60071-2 indicates the correlation between insulation levels and the minimum phase-to-ground air clearances. These are dictated by electrical considerations only and do not include any provisions for construction tolerances, effect of short circuits, extreme environmental conditions, wind effects, safety of personnel, maintenance, corona effects, etc.

3.1 Phase-to-ground air clearances are based on applicable National Grid Saudi Arabia Basic Insulation / Switching Levels and selected / interpolated whenever necessary from the values indicated in IEC 60071-2, Table A.1 with a margin of five percent (5%).

3.2 The minimum phase-to-ground air clearances for outdoor and indoor installations are indicated in Table 08-1.

4.0 PHASE –TO- PHASE AIR CLEARANCES

4.1 Phase-to-phase faults are not desirable in a power system as these cause severe system disturbances. It is therefore to be ensured that phase-to-phase air clearances are greater that phase-to-ground air clearances.

4.2 With this in view phase-to-phase air clearances are kept ten percent (10%) higher than the corresponding phase-to-ground clearances for operating voltages up to and including 230kV.

4.3 For operating voltage 380kV the phase-to-phase air clearance is kept twenty five percent (25%) higher than the corresponding phase-to-ground air clearances.

4.4 The minimum phase-to-phase air clearances for outdoor and indoor installations are also indicated in Table 08-1.

5.0 OTHER MANDATORY CLEARANCES

5.1 National Grid Saudi Arabia substations are accessible only to qualified and authorized personnel. In cases where the qualified personnel are to work in the vicinity of live parts, the minimum air clearances indicated in Table 08-1 are to be suitably increased as practicable to provide sufficient safety in addition to other clearances for the operation and maintenance personnel. The guidelines provided in IEEE (National Electrical Safety Code) to be considered.

5.2 Accordingly, the mandatory clearances to be adopted shall take into consideration the requirements of the local conditions as applicable. The following verticle and horizontal clearances are required to be maintained for personnel and equipment safety when working in the vicinity of live equipment.

5.2.1 Vertical Clearances

For outdoor substations, the minimum vertical clearance from top of any insulator / bushing to the top of plinths of supporting steel work shall in any case not be less than the phase-to-ground clearance specified in Table 08-1 plus 2440 mm being the height of a human being with arms fully extended upwards.

For indoor substations refer Table 08-2

5.2.2 Horizontal Clearances

For outdoor substations, the minimum horizontal safety and maintenance clearance shall be 1700 mm in addition to the minimum phase-to-ground clearance specified in Table 08-1.

5.3 Table 08-2 indicates the minimum vertical clearances to be provided.

5.4 Table 08-3 indicates the Recommended Spacing When Disconnecting Switches are used in Substations

5.5 Table 08-4 indicates the minimum equipment to equipment clearance is same phase in outdoor.

5.6 Table 08-5 indicates the minimum vertical and horizontal maintenance clearances between live part and ground as per clause 5.2.1 and 5.2.2.

5.7 Minimum clearances between panels and wall shall be as specified in TES-P-119.19.

Table 08-1
Minimum Air Clearances in Air Insulated Substation
(Safety and Maintenance Clearance NOT included)

System Nominal Voltage (kV)	Indoor			Outdoor		
	BIL (kV_{peak})	Phase-to-Ground (mm)	Phase-to-Phase (mm)	BIL (kV_{peak})	Phase-to-Ground (mm)	Phase-to-Phase (mm)
11	95	170	190	110	200	220
13.8	95	170	190	110	200	220
33	170	335	370	200	400	440
34.5	170	335	370	200	400	440
69	325	600	730	350	720	790
110	550	**	**	650	1365	1500
115	550	**	**	650	1365	1500
132	650	**	**	750	1575	1730
230	950	**	**	1050	2260	2500
380*	1300	**	**	1425	3600	4500

Note: The clearances given in above table are dictated by electrical considerations / insulation coordination only. At lower voltage upto 200 kV BIL system an increased value of clearance to 500 mm (phase to ground) and 550 mm (phase to phase) is recommended for outdoor applications to avoid flashover from birds and animals.

* For 380kV substations, the clearances are based on the BSL of 1050 kVpeak.

** Indoor air insulated substations rated 110 kV, 115 kV, 132 kV, 230 kV and 380 kV are not normally specified by National Grid Saudi Arabia.

Table 08-2
Minimum Air Clearances for Substations
(Including opertion, safety and maintenace clearances as applicable)

System Nominal Voltage (kV)	outdoor Vertical air clearances		Indoor Vertical air clearnces	
	Between Overhead conductor or live part and ground for personnel safety (mm)*	Between Overhead conductor or live part and asphalt service road inside S/S (mm)	Between Overhead conductor or live part and aisle / passageway (mm)	For bus supports / circuit without disconnecting switch (mm)
11	Not applicable – only factory assembled Indoor metal enclosed switchgear is used in National Grid Saudi Arabia			
13.8	Not applicable – only factory assembled Indoor metal enclosed switchgear is used in National Grid Saudi Arabia			
33**	2900	6000	3350	335
34.5	Not applicable – only factory assembled Indoor metal enclosed switchgear is used in National Grid Saudi Arabia			
69	3200	6200	3650	660
110	3810	6800	Indoor Air Insulated 110 kV, 115 kV, 132 kV, 230 kV and 380 kV substations are not normally specified by National Grid Saudi Arabia	
115	3810	6800		
132	4020	7100		
230	4700	7900		
380	6050	9700		

* The height indicated here is up to plinth / foundation level of supporting steel work

** These clearnaces apply where indoor metalclad switchgear is not used.

Table 08-3
(Recommended Spacing When Disconnecting Switches are used in Substations)

System Nominal Voltage (kV)	Recommended Spacing (Center to Center)	
	For vertical break disconnect switches, power fuses and bus supports (mm)	For horizontal break disconnect switches and expulsion type fuses (mm)
11	Not applicable – only factory assembled indoor metal enclosed switchgear is used in National Grid Saudi Arabia	
13.8	Not applicable – only factory assembled Indoor metal enclosed switchgear is used in National Grid Saudi Arabia	
33*	914	1220
34.5	Not applicable – only factory assembled Indoor metal enclosed switchgear is used in National Grid Saudi Arabia	
69	1520	1830
110	2130	2740
115	2440	3350
132	2740	3960
230	3960	5500
380	5000	7400

Note: Above are recommended spacing – actual are to be based on manufacturer's data. However, designer to ensure that minimum phase-to-phase and phase-to-ground clearances as specified in Table 08-1 are maintained.

* These spacing apply where indoor metal enclosed switchgear is not used.

Table 08-4
(Outdoor - Minimum equipment to equipment clearance in same phase)

110 kV / 115 kV / 132 kV	230 kV	380 kV
2500 mm	3500 mm	4000mm

Table 08-5
(Minimum phase to ground maintenance clearance
a per Cl: 5.2.1 & 5.2.2 for outdoor substations)

System Nominal Voltage (kV)	Vertical clearance (mm)	Horizontal clearance (mm)
11	2640	1900
13.8	2640	1900
33	2900	2100
34.5	2900	2100
69	3200	2420
110	3810	3065
115	3810	3065
132	4020	3275
230	4700	3960
380*	6050	5300

6.0 BIBLIOGRAPHY

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6. IEC 60071-2: "Insulation Coordination, Part 2: Applications Guide"

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